

Audubon Nature Institute's White Alligators

The American Alligator (*Alligator mississippiensis*) has been displayed in American zoological parks since the late 19th century. The earliest institutional record of a zoo specimen in the United States appears to be to a female acquired by the Philadelphia Zoo in 1873 (ZIMS 2020); however, the species was kept even earlier than this at some of the country's early zoo precursors including the Central Park Menagerie in New York. In European zoological parks, *A. mississippiensis* has been displayed even longer, with possibly the earliest known specimen maintained at the Tower of London menagerie sometime prior to 1829 (Bennett 1829). The Jardin des Plantes of the Muséum National d'Histoire Naturelle in Paris acquired several live specimens from a New Orleans collector in 1852 (Murphy 2007), and the London Zoological Gardens acquired several specimens between the late 1840s and 1850s (Sowerby 1857; Anonymous 1862). Today, *A. mississippiensis* is the most ubiquitous crocodilian displayed in zoological parks and related facilities throughout the world (Ziegler et al. 2017).

As a species, *A. mississippiensis* is no stranger to zoo-goers in the United States—out of the country's several hundred indigenous reptile species, it is probably the most infamous and recognizable to the American layperson. The species also represents one of the country's greatest wildlife conservation success stories, having rebounded from decreased population numbers thanks to rigorous scientific study, state and federal protections, and management programs that relied on sensible sustained-use and harvest practices (Joanen et al. 1997; Louisiana Department of Wildlife and Fisheries [LDWF] 2019).

In Louisiana, *A. mississippiensis* is distributed throughout the entirety of the state, inhabiting most freshwater bodies from lakes and rivers to roadside ditches and coastal marshes. The species bears strong cultural significance in the state, especially in Cajun communities where alligators are deeply entrenched in the cuisine, traditions and folklore of its people (Glasgow 1991). Additionally, commercial alligator ranching for leather production (with meat produced as a valuable by-

product) has become a robust industry in Louisiana (Joanen et al. 1997), which is now the largest producer of alligator products in the country. As of 2018, there were more than 50 commercial alligator farms operating throughout the state producing hides and meat products valued at more than \$100 million annually (LDWF 2019). Ecotourism focusing heavily on sightseeing wild alligators has also become an important industry in rural areas, with numerous "swamp tour" companies operating throughout the state. Because of these strong cultural, socioeconomic and ecological ties to the state, *A. mississippiensis* has been a quintessential display species for zoological parks in Louisiana and can also be viewed at various other facilities including nature centers, wildlife refuges, and sanctuaries.

In New Orleans, Louisiana, alligators have been an important fixture at Audubon Zoo throughout much of its 100+ year history, dating back to at least 1924. While alligators consistently rank among the most popular and sought out species by visitors to Audubon Zoo, the arrival and exhibition of a group of white alligators in 1987 brought unprecedented attention and interest in the species. These individuals, together with several additional white specimens acquired by the Audubon Nature Institute since then have now captivated local and global audiences for more than three decades and have become cultural icons in New Orleans and surrounding areas of southern Louisiana. The following account provides a brief overview of herpetology at the Audubon Nature Institute and details the history and origins of these unique white specimens, noteworthy aspects of their captive management, and their rise to cultural phenomenon status both within the Institute and abroad. For earlier popular accounts on the Audubon Nature Institute's white alligators, see Walls and Walls (1991) and Armand and James (1994).

HERPETOLOGY AT THE AUDUBON NATURE INSTITUTE

Audubon Zoo was founded in 1916 on a 14-acre plot of land within Audubon Park, located in uptown New Orleans. The property's footprint was later expanded to 58 acres, which it remains to this day. Like many other municipal zoos around the same time (e.g., Mendyk and Rost 2018), Audubon Zoo experienced major financial hardships through the 1950s, 60s, and 70s. By the late 1970s, a new not-for-profit organization assumed management and oversight of many of the facility's operations from the city and would eventually play a pivotal role in reviving and transforming the struggling facility (Forman and Logsdon 1985). Later, through its new moniker—the Audubon Nature Institute [ANI] (established in 1989; not to be confused with the ornithologically-focused Audubon Society), the organization would go on to develop several additional zoological facilities within the city of New Orleans including the Audubon Aquarium of the Americas (1990), Freeport-McMoRan Audubon Species Survival Center (1993), Audubon Louisiana Nature Center (1994), and Audubon Butterfly Garden and Insectarium (2008). In addition to its zoological facilities,

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ANI also manages several public parks and recreational spaces throughout the city of New Orleans. Combined annual visitation to the Institute's various zoological facilities averages 1.6 million people.

Based on available institutional records, at least 398 species of herpetofauna have been maintained in ANI facilities to date. As far as it can be determined from an early Audubon Park commission report, it appears that reptiles have been maintained as early as 1924, with various species displayed in an expanded section of the zoo's Odenheimer Aquarium complex around 1925–1926. A small building for Galapagos Tortoises (*Chelonoidis* spp.) was also constructed around this time. As part of several Works Progress Administration projects, an additional tortoise yard and snake house were completed by the end of 1935, and new turtle enclosures and an alligator pool were constructed in 1936 (Stern 1984). The vast majority of reptiles and amphibians displayed at Audubon Zoo to date, however, have been maintained over just the last four decades. Opened in 1984, the zoo's Louisiana Swamp exhibit complex has displayed an assortment of native Louisianan herpetofauna, with *A. mississippiensis* serving as a focal species throughout the complex. An official Herpetology Department (separate from the Louisiana Swamp Department) was established in 1986, just prior to the opening of the zoo's Reptile Encounter building the following year (Bowler 1989), which led to a dramatic increase in the size and scope of the zoo's herpetological collection. As of October 2020, the zoo's herpetological collection comprised a total of 597 individual specimens representing 100 reptile and 25 amphibian species that are mostly maintained by its Herpetology Department (ca. 90%), but also distributed across its Louisiana Swamp, Animal Ambassador, and Asian Domain Departments. Alligators remain a prominent fixture in the zoo's collection, with a total of 18 *A. mississippiensis* of various age groups currently displayed in several indoor and outdoor areas throughout the Louisiana Swamp exhibit complex. Additional juvenile specimens are also usually maintained off-exhibit for use in educational outreach.

In downtown New Orleans, the Audubon Aquarium of the Americas has maintained at least 36 species of herpetofauna since its opening in 1990. Its current herpetological collection comprises a total of 15 reptile and 8 amphibian species, including *A. mississippiensis*, that are maintained on display and in off-exhibit areas for educational programming. Located just a few blocks away from the aquarium until recently, the Audubon Butterfly Garden and Insectarium has displayed several species of native herpetofauna including juvenile *A. mississippiensis*; this facility will soon be relocated to and incorporated into the aquarium. The Audubon Louisiana Nature Center, located in New Orleans East, has also maintained several reptiles and amphibians for use in educational programming including *A. mississippiensis*.

Since the late 1980s, staff members from Audubon's zoo, aquarium, and nature center have authored many publications on a broad range of herpetological topics and taxa including *A. mississippiensis* (Thomas 1999; Wilson et al. 2006; Mendyk and Terrell 2019). These include works on natural history (Mendyk et al. 2018a; Mendyk and Terrell 2019), systematics and biogeography (Vaughan et al. 1996; Thomas 1997; Thomas and Fernandes 1996; Thomas and Ineich 1999), conservation (Hinkson et al. 2020), captive management (Davis 2006; Mendyk 2018a; Murphy et al. 2019; Hartdegen and Mendyk 2019a,b; Peeling and Mendyk 2019), reproduction (Litton et al. 2018),



FIG. 1. Leucistic and normal-colored sibling *Alligator mississippiensis* collected near Houma, Terrebonne Parish, Louisiana in 1987.

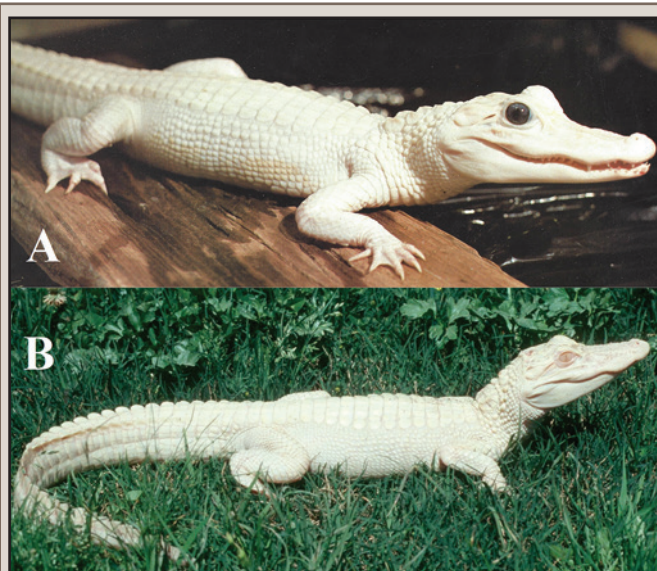


FIG. 2. Comparison of eye pigmentation in leucistic (A) versus amelanistic (albino) (B) *Alligator mississippiensis*.

behavior (Chiszar et al. 1985; Pramuk and Hiler 1999; Mendyk 2018c, 2020a; Mendyk et al. 2018b; Murphy et al. 2019), growth and development (Bjorndal et al. 2001), longevity (Snider and Bowler 1992), genetics (Jimenez-Mena et al. 2016), veterinary medicine (Jacobson et al. 1992; Miller et al. 1997, 2003; Stacy et al. 2017; Wilson et al. 2006), snakebite management (Perry-Richardson 1990; Field et al. 2016), invasive species (Heckard et al. 2013; Glorioso et al. 2018; Pellecchia et al. 2019), zoo history (Mendyk and Rost 2018, 2020b), exhibit design (Snider 1988; Bowler 1989), staff development (Murphy et al. 2020), and biographical and bibliographic research (Mendyk 2018b, 2019a,b, 2020c). Several additional publications from zoo staff are forthcoming (Fink et al., *in press*; Mendyk and Augustine, *in press*; Mendyk and Block, *in press*; Mendyk and Warwick, *in press*).

HISTORY AND ORIGINS OF AUDUBON'S WHITE ALLIGATORS

In late August 1987, a recently hatched nest of white alligators was discovered by a fisherman in swamplands near Houma, Terrebonne Parish, Louisiana. Nine white hatchlings were initially collected by the fisherman, who then distributed some

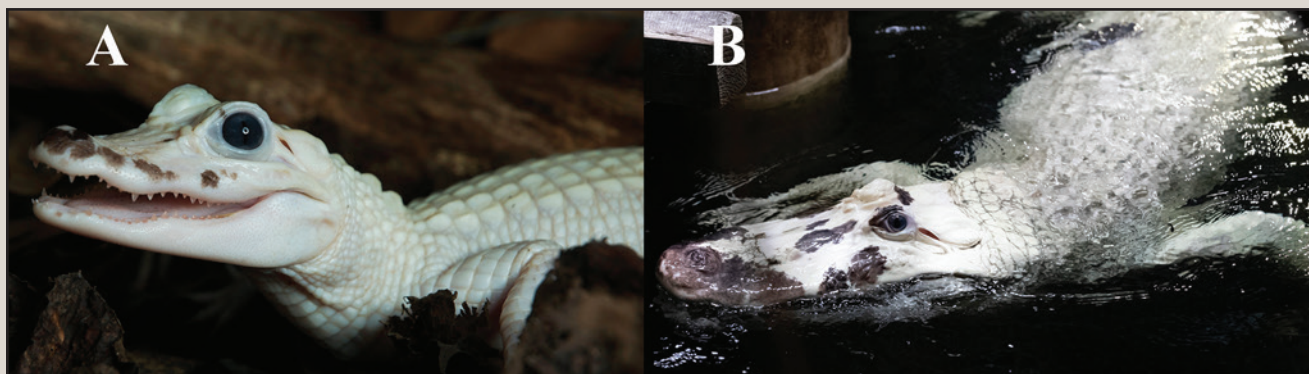


FIG. 3. Ontogenetic changes in facial markings in a leucistic *Alligator mississippiensis* as a juvenile (A) and adult (B). Photographs courtesy of Audubon Nature Institute.



FIG. 4. Leucistic *A. mississippiensis* on display in Audubon Zoo's Louisiana Swamp Exhibit, early 1990s.

among friends (including a gentleman in St. Louis) and gave six animals to Audubon Zoo.

Upon learning about the collection of these specimens, staff from the Louisiana Land and Exploration Company (hereafter LLE), an oil and gas exploration and refinery company which owned the wetlands property on which the nest was found, recovered several additional white hatchlings over the course of several months along with seven normal-colored siblings (Fig. 1), bringing the total number of white individuals collected at this site to 19.

Estimated to be around one to two weeks old at the time of their discovery and measuring 24.2–26.0 cm in total length, the white hatchlings exhibited leucism, a rare genetic mutation resulting in the absence of melanocytes in the epidermis (Larson and Cardeilhac 1996). Unlike albino, or amelanistic alligators which exhibit pink eyes due to their complete lack of iris pigmentation, these leucistic individuals possessed dark blueish eyes (Fig. 2). Aside from their snow-white background color, many of these specimens also featured small patches of dark coloration on the head, neck, and/or dorsum that increased in size with age (Fig. 3).

With the exception of the single individual sent to St. Louis which died shortly thereafter, all other leucistic individuals collected by the fisherman were eventually recovered by LLE, who at the time was already licensed by the state to maintain

captive alligators for commercial rearing purposes. Four leucistic individuals and six normal-colored siblings were officially donated to Audubon Zoo by LLE in the fall of 1987 for display in its Louisiana Swamp exhibit complex (Fig. 4). Two of these leucistic specimens were later transferred to Audubon's Aquarium of the Americas when the facility opened in 1990. Ownership of the remaining 14 leucistic individuals was retained by LLE, who reared them at its alligator farm in Golden Meadow, Louisiana (established in 1983) as well as at an additional site near Des Allemands, Louisiana where the company oversaw a marsh conservation project. Several individuals from this group were transferred back and forth between LLE and ANI facilities over the next several years, and one specimen was loaned out to the Houston Zoo for long-term exhibition. At LLE's two properties, breeding groups comprised of leucistic males and normal-colored females were assembled with the goal of commercially producing leucistic offspring (see below). Two leucistic animals died at the Golden Meadow farm, including the only female from the group which was killed by a larger leucistic sibling.

Aside from the animals that remained on display in New Orleans at Audubon's zoo and aquarium and the specimen on loan to Houston Zoo, several of the leucistic individuals from this cohort were loaned out to other institutions for temporary exhibition through ANI between the late 1980s and early 2000s. This included more than 30 zoological facilities in the United States, as well as Mexico and Japan.

When LLE merged with Burlington Resources Inc. in 2001, the company decided to discontinue its alligator farming operations and donated its remaining 12 leucistic alligators (one of which was still on loan to Houston Zoo) and five normal-colored, potentially heterozygous offspring hatched at its facilities (see below) to ANI. To determine a value for the group being transferred and when compared against the going market price of amelanistic alligators at the time and considering the rarity of leucistic specimens and their exhibition value to zoological parks, adult leucistic individuals were professionally appraised at \$450,000 each, down from \$500,000–700,000 just a few years earlier in 1993 likely due to the increased availability of amelanistic individuals which now filled a similar "novelty" niche in zoological parks and related facilities. Construction of a new facility for housing these additional specimens was planned for a behind-the-scenes area of the zoo, but these plans were eventually abandoned. These additional animals were maintained at the Golden Meadow property until they could all be transferred to other zoological facilities including Omaha's Henry Doorly Zoo (2006), Palm Beach Zoo (2007), and Gatorland



FIG. 5. "Two-spot", a 33+ year-old male leucistic *Alligator mississippiensis* at Audubon Zoo.

(2008) Today, one individual from this original cohort remains with ANI and is on display at Audubon Zoo (Fig. 5).

In November 1994, a ca. 40 cm long leucistic alligator was donated to ANI by an anonymous individual who had purportedly collected it a few months earlier near Venice, in Plaquemines Parish, Louisiana. Because state legislation prohibiting the collection of white alligators from the wild had been enacted a few years earlier in response to the collection of the original group of leucistic hatchlings in 1987, ownership of this specimen was assumed by the Louisiana Department of Wildlife and Fisheries (LDWF), who then loaned the specimen to the Audubon Zoo for display. As with the previous leucistic specimens, this individual also exhibited some small patches of dark coloration on the head and was later determined to be female. It was maintained at the zoo until 1999 when it was transferred back to state officials and died later that year after being displayed for educational and promotional use at leather and fur trade shows.

In 2009, ANI acquired two yearling leucistic alligators from LDWF that were collected by officials in the Mandalay National Wildlife Refuge near Houma, in Terrebonne Parish. These two specimens, which also featured some small dark patches of pigmentation and later proved to both be males,

measured around 33 cm in total length and were found together with normal-colored siblings. As juveniles, they were initially displayed at Audubon's Butterfly Garden and Insectarium before being transferred to other ANI facilities as they grew larger. One of these individuals was transferred to the North Carolina Aquarium on Roanoke Island in 2013; the other currently resides at Audubon's Aquarium of the Americas.

Most recently in 2019, Audubon Zoo received a ca. 150 cm long male leucistic alligator from LDWF. This animal had been collected as a yearling near Houma in 2011, and since then had been kept as an educational specimen at the Rockefeller Wildlife Refuge. This specimen also exhibits some small patches of dark pigmentation which have become more prominent with age and is currently on display at Audubon Zoo (Fig. 6).

CAPTIVE MANAGEMENT

Husbandry and Reproduction.—Animals from the 1987 cohort reared at the zoo and aquarium grew considerably slower than their leucistic siblings raised at LLE's facilities. At around five years in age, there was a marked difference in body sizes between siblings reared at each facility, ranging between 1.5 m and 22–27 kg, and 2.4 m and 113 kg (Burnette 1994), with the smaller individuals residing at the zoo and aquarium, and larger individuals at LLE's facilities. This disparity in growth was most likely attributed to differences in husbandry practices and rearing conditions between the facilities. For example, at least some individuals at the farm were reared using techniques and conditions developed for commercial alligator production that favored rapid growth rates, such as maintaining them in indoor heated pools maintained at a constant temperature of 31°C and feeding a diet almost entirely comprised of nutria supplemented with vitamins and minerals (Larsen and Cardeilhac 1996). At the zoo and aquarium, rapid growth rates were not necessary or desirable, so animals were fed less and experienced more natural fluctuations and seasonality in temperatures, thereby contributing to more typical growth rates for the species (e.g., Chabreck and Joanen 1979; Rootes et al. 1991). A comparative study of growth rates between leucistic individuals and their normal-colored siblings had been initiated in collaboration with outside researchers; however, we could find no record of any resulting publication.

Whereas ANI initially chose not to attempt captive breeding



FIG. 6. Subadult leucistic *Alligator mississippiensis* acquired by Audubon Zoo in 2019.



FIG. 7. Outdoor-housed leucistic *Alligator mississippiensis* exhibiting sunburn; mid to late 1990s.

with their leucistic alligators (but later reconsidered this as a possibility [D. Maloney, pers. comm.]), animals at LLE's farm were reared specifically for this purpose. Prior to its death in 2000, the only leucistic female from the original group unexpectedly produced eggs in 1998, but they were scattered in the pool and non-viable. Between 1994 and 1995, three seven-year-old leucistic males at LLE's farm were tested for their production of spermatozoa and to determine whether sufficient quantities could be collected for artificial insemination attempts (Larsen and Cardeilhac 1996). Only two of these individuals produced sperm, and despite attempts with artificial insemination in six normal-colored females in 1994 and 1995, only a single clutch of infertile eggs was produced. In 1995, two normal-colored females that had been paired with leucistic males at the Golden Meadow farm produced eggs, but only one of the clutches contained fertile eggs (8 out of 42). Presuming that leucism in alligators is the result of an autosomal recessive mutation as has been observed in other reptiles, two normal-colored, heterozygous offspring hatched from this clutch in 1995. The following year, a normal-colored female at the Des Allemands site that had been paired with a leucistic male produced 28 fertile eggs (out of a clutch of 35) which resulted in an additional five normal-colored offspring that were incubated at temperatures to produce females. LLE discontinued its alligator farming operations before it could breed these heterozygous individuals back to the original leucistic males and therefore never succeeded in producing leucistic offspring; additionally, no efforts were made by ANI to breed the five suspected heterozygous offspring it received from LLE with the remaining group of leucistic adults in 2001. These offspring were eventually transferred to Gatorland, where females from this group are currently paired with leucistic males from the original 1987 cohort for breeding (M. McHugh, pers. comm.). The marginal reproductive success experienced by LLE led some people involved with the project to suspect that their genetic condition was responsible; however, leucistic lineages of many other reptile species have successfully been reproduced in captivity without difficulty (e.g., *Python regius*, *Malayopython reticulatus*, *Epicrates maurus*, *Pantherophis obsoletus*), casting doubts on such a claim. Captive diets (see Lance et al. 2001) and general husbandry practices may have played a greater contributing role.

Morbidity, Mortality, and Lifespans.—According to Larsen and Cardeilhac (1996), leucistic alligators from the original 1987 cohort did not appear to possess the same congenital defects or

complications commonly seen in amelanistic animals such as low immunocompetence, fibrotic internal organs, poor vision or poor hearing, and had serum globulin levels similar to normal-colored individuals. Some behavioral differences in feeding response and basking activity were observed between leucistic and normal-colored individuals at the zoo but may have been related to differences in rearing conditions, with leucistic animals typically housed individually indoors under artificial lighting, and normal-colored individuals housed outdoors with other individuals (I. Wyatt, pers. comm.). The single leucistic female acquired in 1994 experienced chronic health issues for much of its life including ulcerative bacterial dermatitis and periods of weight loss, and only lived a few years (Barnett et al. 1999).

In addition to the animals maintained at its own zoological facilities, ANI also provided veterinary care for the leucistic specimens owned and maintained by LLE at its breeding farms. Several chronic health issues were seen in individuals from Audubon's and LLE's collections including skin issues (some of which were associated with viral papillomatosis) and *Mycobacteria* infections, and at least one individual developed bilateral cataracts (J. Torres, pers. comm.). Most notably, many of these individuals were treated for lead poisoning and some for zinc toxicosis, attributed to the ingestion of lead pellets used to hunt the wild nutria that comprised the bulk of their diets (see also Camus et al. 1998) and consumption of coins thrown into their exhibit by aquarium visitors, respectively. Treatment required manual and endoscopic removal of the metallic foreign bodies from the animals' gastrointestinal tracts as well as chelation therapy. To address the chronic issue of lead consumption, alligators at the zoo were at one point switched to a diet comprised of human consumption-grade nutria meat. Today, alligators at Audubon Zoo are offered Mazuri® crocodilian pellets, whole fish (mackerel, trout), chicken, quail, rats, and horse meat; at the Audubon Aquarium of the Americas, they receive a varied diet comprised of assorted types of fish, rodents, poultry, and crocodilian-formulated pellets and gels (D. Murphy, pers. comm.).

Leucistic alligators present challenges for their maintenance in captivity due to their sensitivity to ultraviolet light on account of their lack of skin pigmentation. Animals housed outdoors at both Audubon Zoo and LLE's Golden Meadow facility experienced chronic sunburns (Fig. 7), which led the zoo to move their leucistic animals indoors under artificial lighting. On the other hand, periodic health screenings of leucistic individuals reared indoors at the Golden Meadow farm revealed that individuals exhibited dangerously low 25-Hydroxyvitamin D plasma levels, a likely consequence of insufficient UV exposure, suggesting that some intermediate level of sunlight exposure is needed.

Confirmed sources of mortality among captive leucistic alligators have included clostridial infection (*Clostridium* sp.), lead poisoning, zinc toxicosis, pneumonia and heart disease, bacterial enteritis, and trauma inflicted by a conspecific. As of October 2020, seven of the original 19 leucistic individuals collected in 1987 are still living and now over 33 years old. For the nine deceased individuals from this group with confirmed death dates, their average lifespan was 17.3 ± 6.3 years (range: 11.4–28.2 years).

Miscellany.—Leucistic alligators often change color in varying degrees from pure white to a “blushing” pinkish color when handled or disturbed and during transport. This change, which can be especially apparent in the head, neck, feet and tail base regions, has been observed in both juvenile and adult individuals and is probably associated with increased blood flow to the



FIG. 8. “Piebald” (top) and leucistic (bottom) *Alligator mississippiensis* hatched at a Florida alligator farm in 2014. Photograph courtesy of Nathan Sweeting.

vessels closest to the surface of the skin during stress or exertion.

ADDITIONAL CASES OF LEUCISM IN *ALLIGATOR MISSISSIPPIENSIS*

Leucism has been reported in a few other crocodilian species including *Crocodylus porosus* (Bustard and Kar 2019) and *Caiman crocodilus* (Chavez 2006). However, when reviewing photographs accompanying these accounts, it is evident that these individuals did not exhibit the snow-white body coloration and dark blue eyes typical of the leucistic alligators discussed herein and in other leucistic reptiles maintained in captivity. Instead, the body coloration of those individuals has been pale yellow or light beige with normal, albeit faded body markings, which is more in line with what has been referred to in other reptiles as “hypomelanism” or the reduction of melanin in the skin—rather than a total absence thereof (e.g., Ihle et al. 2000; Turner 2011)—and is similar to several *C. porosus* specimens depicted and discussed by Campbell and Winterbotham (1985).

Leucism and albinism in alligators appear to be rarely occurring genetic aberrations. Some amelanistic specimens have been sourced from the wild (Allen and Neill 1956) and hatched in alligator farms (see e.g., Berry 1995; Irwin 2010), and amelanistic offspring are now commercially produced by reptile hobbyists and breeders (G. Lepera, pers. comm.). Despite the apparent rarity of leucism in wild alligator populations, several additional confirmed and suspected cases have been recorded from the southeastern United States over the past 130+ years that are noteworthy. Some of these, however, are from anonymous sources in old newspapers which cannot be verified, and since most of these accounts make no reference to eye color, it is not possible to truly ascertain whether they reference leucistic or amelanistic individuals.

Alabama.—An early 1886 news note briefly reported the discovery of a “milk white” alligator in a pond in Escambia, Alabama along with one that was “parti-colored” (= mixed colors) (Anonymous 1886). From these limited descriptions, the former animal may have been leucistic, whereas the latter may refer to an individual with the same condition but with areas of normal pigmentation as observed in many of Audubon’s leucistic alligators (Fig. 2), or possibly even a “piebald” individual (see e.g., Fig. 8).



FIG. 9. Artwork depicting leucistic alligators is prominently displayed throughout the Audubon Nature Institute’s administrative offices.

Florida.—In 1928, the Florida Alligator and Ostrich Farm in Jacksonville acquired a pure white alligator that was collected somewhere in Florida (locality unknown). The following year, the farm was in negotiations with the Zoological Society of London to sell the specimen for US \$2,000, with the Society purportedly sending a representative to the farm to examine the specimen in person (Anonymous 1929a,b); its final disposition is unknown. A leucistic alligator was hatched at a Florida alligator farm in 2014, and briefly displayed at the Smooth Waters Wildlife Park in DeLeon Springs (N. Sweeting, pers. comm.). Like previous leucistic alligators, this specimen also exhibited dark blue eyes and some patches of dark pigmentation (Fig. 8). Its current whereabouts is unknown.

Louisiana.—In addition to the 23 aforementioned leucistic specimens collected in Terrebonne and Plaquemines Parishes between 1987 and 2011, Audubon Zoo was contacted by an anonymous individual in February 2019 about a meter-long white alligator that had reportedly been spotted on his property, also in Terrebonne Parish. However, as this lead was not pursued by ANI, the claim could not be verified. Several other unsubstantiated reports of white alligators have been received by the zoo since the early 1990s (D. Maloney, pers. comm.). While there have been various amelanistic specimens hatched in Louisiana alligator ranches from eggs collected in several parishes within the state (Berry 1995; R. Elsey, unpubl. data), there have been no confirmed reports of leucistic alligators hatching at ranches. The fact that three separate cohorts of leucistic alligators have been recorded from a fairly restricted area in Terrebonne Parish (ca. 5390 km²) over a 20+ year period is peculiar, especially since no other leucistic specimens—besides the single specimen said to have been collected near Venice in 1994—have been documented from anywhere else in the state over the last several decades despite extensive statewide harvesting of wild alligator eggs for commercial ranching efforts each year. The discovery of both normal and leucistic individuals from the same wild clutch in 1987 might suggest a natural occurrence of recessive alleles for this trait within the population, or possibly even a case of multiple paternity (e.g., Davis et al. 2001; Lance et al. 2009). Comparative genetic analyses between wild-caught captive leucistic specimens and normal-colored individuals from the same areas could make for an interesting study and potentially shed light on this apparent phenomenon.

Mississippi.—A large white alligator was purportedly spotted regularly at a property in Yazoo City over the course of several years in the 1890s (Anonymous 1896), where it allegedly attacked and predated several small calves. It was also reported that normal-colored alligators were once commonly encountered in the area

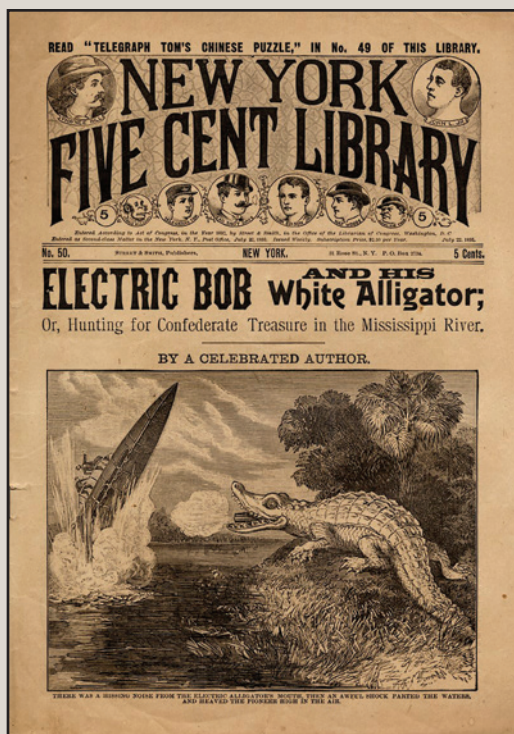


FIG. 10. Cover page of Robert T. Toombs' (1893) *Electric Bob and his White Alligator*, an early science fiction dime novel. The work satirizes other 'young inventor' novels of the era such as those by Frank Reade Jr. and Jack Wright, with the New Orleanian protagonist inventing a seaworthy submersible vehicle disguised as a white alligator.

but had been displaced or disappeared upon the arrival of the white specimen, and that it had evaded several hunting parties that targeted it. In 2000, Audubon Zoo was contacted by a property owner in Gautier about a white alligator that had been seen in the bayou behind a house near the Pascagoula River. The authenticity of these claims could not be confirmed; however, state officials were also apparently aware of several reported sightings of white alligators in the area at the time.

South Carolina.—In September 2003, South Carolina Department of Natural Resources officials seized three leucistic hatchling alligators from a group of men affiliated with a local serpentarium on Edisto Island who had purportedly collected them from a wild nest on the grounds of a resort on Hilton Head Island earlier that month (Anonymous 2006a). All three specimens were in poor physical condition, and subsequently transferred by officials to the Riverbanks Zoo (Pfaff 2005; S. Pfaff, pers. comm.). Two died within 11 days of their seizure from yolk peritonitis (S. Pfaff, pers. comm.); the remaining individual was cared for by the zoo while the ensuing legal case was carried out and eventually displayed publicly in 2006 (Anonymous 2006b). It had experienced chronic health issues and died later that year from an apparent intestinal infection (Santaella 2006). In 2014, a pod of hatchling alligators containing several leucistic individuals was discovered in coastal South Carolina (Rainwater et al. 2020). These animals were subsequently collected by US Fish and Wildlife Service officials and treated by veterinarians for malnutrition. Surviving individuals were transferred to several aquariums and wildlife centers in the southeastern US for display; however, all of them failed to live more than a few years (Rainwater et al. 2020).

Unknown origin.—A "milk white" alligator was acquired by and displayed at a small menagerie at Pacific Park, in Long Beach, California in 1907 (Anonymous 1907). On the same morning that Ringling Brothers Circus offered \$500 for the specimen, it was stolen from the park. With little understanding of genetic aberrations at the time, it was believed that a disease was responsible for the animal's unusual body coloration.

WHITE ALLIGATORS AS CULTURAL ICONS

It is unlikely that anyone could have predicted the level of interest, sensationalism and international media attention that Audubon's white alligators would go on to generate. At the organizational level, their popularity at Audubon Zoo and Audubon Aquarium of the Americas grew quickly and led to their embodiment as the unofficial mascot of the Institute. While there have been other iconic associations between zoological institutions and flagship species in their collections such as Sea World and Killer Whales (*Orcinus orca*) and the Smithsonian National Zoological Park and Giant Pandas (*Ailuropoda melanoleuca*), as far as we can tell, Audubon's close association with its white alligators may very well be the only major North American zoological institution represented in such a capacity by a reptile. Various depictions and artistic representations of white alligators are prominently featured throughout ANI's zoological facilities, including an 8.5 m-tall white alligator statue in the zoo's water park, large white alligator 'totems' in the zoo's gift shop, and various paintings and artwork displayed throughout the organization's administrative offices (Fig. 9).

In New Orleans and some of the surrounding parishes in southern Louisiana, white alligators have become cultural icons and are frequently depicted in local artwork, fanfare and public displays. Interestingly, the appearance of white alligators in fictional literary accounts of the region precedes the discovery of Audubon's original 1987 group of leucistic hatchlings by nearly a century (Toombs 1893; Fig. 10). In more recent works, white alligators have been nicknamed "swamp ghosts" and are featured in fictional novels and children's books as well as posters and murals by local artists (Figs. 11, 12). In Morgan City, St. Mary Parish, ca. 135 km southwest of New Orleans and just west of the various sites in Terrebonne Parish where leucistic alligators have been collected, a seasonal Christmas holiday display first erected in 2018 greets visitors to the town with a trio of blue-eyed leucistic alligators (in place of reindeer) in harnesses pulling Santa's "sleigh"—a shrimp vessel (Fig. 13).

Beyond their display in New Orleans at ANI's zoological facilities and the many other zoos and aquariums that have temporarily exhibited them abroad, Audubon's white alligators have reached millions of additional people through various media appearances. These have included appearances on popular American television programs such as the Late Show with David Letterman, the Tonight Show, and the Today Show, and through media coverage by major news networks including CNN and CBS. For many years, practically any news about these leucistic individuals generated widespread media attention, contributing to probably hundreds of articles written about them that were subsequently picked up by news agencies around the world.

CONCLUDING REMARKS

Audubon Nature Institute's leucistic alligators have been an important fixture of its various zoological facilities for more



FIG. 11. Local poster artwork depicting leucistic alligator.

than 30 years, and it would be difficult for its visitors and staff to envision Audubon's zoo or aquarium without these iconic specimens. The future of such herpetological "oddities" in zoological parks remains uncertain, however, as more institutions restructure their collections and prioritize exhibit spaces for species of conservation concern. Considering the finite resources and spatial constraints of zoos and aquariums, is it appropriate for facilities to allocate valuable exhibit space and resources to such novelties when they could perhaps better be afforded to species in need of conservation support? We fully recognize the complexity of the issue where further discussion of this topic will be crucial as



FIG. 12 White alligator mural on VW Bus. French Quarter, New Orleans.

zoos continue to reform and refine their conservation priorities and living collections, but it is beyond the scope of this article.

Acknowledgments.—We thank Daine Appleberry, Donna Bear, Erin Borgardt, Kevin Bowler, Shane Boylan, Vickie Edel, Jessica Foti, Trudy Ehrich, Joel Hamilton, Remy Lazare, Zack Lemann, Greg Lepera, Melanie Litton, Dan Maloney, Dee Murphy, Scott Pfaff, Thomas Rainwater, Carmen Rucker, Andy Snider, Nathan Sweeting, Jamie Torres, GingerVanGeffen, Jonathan Vogel, and Isaac Wyatt for providing useful information, anecdotes, and various other courtesies. Bob Thomas, Rolf Larsen, Joel Hamilton, and the Smithsonian Institution Libraries provided literature and bibliographic information, Jonathan Vogel, Lee Romaine, Nathan Sweeting, and Ian Kramar generously provided photographs, and Kevin Bowler, Judith Block, and James Murphy offered useful feedback on an earlier draft of this manuscript. This article is dedicated to Alan F. Rost (1949–2019), former Biological Programs Registrar of the Jacksonville Zoo and Gardens and lifelong student and fancier of zoo history.



FIG. 13. A Christmas display featuring leucistic alligators in place of Santa's reindeer greets visitors to Morgan City, Louisiana during the holiday season.

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